



Reading: Glossary terms from course 2, module 3

New terms and their definitions: Course 2 Module 3

ACK flag: One of the TCP control flags. ACK is short for acknowledge. A value of one in this field means that the acknowledgment number field should be examined

Acknowledgement number: The number of the next expected segment in a TCP sequence

Application layer: The layer that allows network applications to communicate in a way they understand

Application layer payload: The entire contents of whatever data applications want to send to each other

CLOSE: A connection state that indicates that the connection has been fully terminated, and that no further communication is possible

CLOSE_WAIT: A connection state that indicates that the connection has been closed at the TCP layer, but that the application that opened the socket hasn't released its hold on the socket yet

Connection-oriented protocol: A data-transmission protocol that establishes a connection at the transport layer, and uses this to ensure that all data has been properly transmitted

Connectionless protocol: A data-transmission protocol that allows data to be exchanged without an established connection at the transport layer. The most common of these is known as UDP, or User Datagram Protocol

Data offset field: The number of the next expected segment in a TCP packet/datagram

Demultiplexing: Taking traffic that's all aimed at the same node and delivering it to the proper receiving service

Destination port: The port of the service the TCP packet is intended for

ESTABLISHED: Status indicating that the TCP connection is in working order, and both sides are free to send each other data

FIN: One of the TCP control flags. FIN is short for finish. When this flag is set to one, it means the transmitting computer doesn't have any more data to send and the connection can be closed

FIN_WAIT: A TCP socket state indicating that a FIN has been sent, but the corresponding ACK from the other end hasn't been received yet

Firewall: It is a device that blocks or allows traffic based on established rules

FTP: An older method used for transferring files from one computer to another, but you still see it in use today

Handshake: A way for two devices to ensure that they're speaking the same protocol and will be able to understand each other

Instantiation: The actual implementation of something defined elsewhere

Listen: It means that a TCP socket is ready and listening for incoming connections

Multiplexing: It means that nodes on the network have the ability to direct traffic toward many different receiving services

Options field: It is sometimes used for more complicated flow control protocols

Port: It is a 16-bit number that's used to direct traffic to specific services running on a networked computer

Presentation layer: It is responsible for making sure that the unencapsulated application layer data is actually able to be understood by the application in question

PSH flag: One of the TCP control flags. PSH is short for push. This flag means that the transmitting device wants the receiving device to push currently- buffered data to the application on the receiving end as soon as possible

RST flag: One of the TCP control flags. RST is short for reset. This flag means that one of the sides in a TCP connection hasn't been able to properly recover from a series of missing or malformed segments

Sequence number: A 32-bit number that's used to keep track of where in a sequence of TCP segments this one is expected to be

Server or Service: A program running on a computer waiting to be asked for data

Session layer: The network layer responsible for facilitating the communication between actual applications and the transport layer

Socket: The instantiation of an endpoint in a potential TCP connection

Source port: A high numbered port chosen from a special section of ports known as ephemeral ports

SYN flag: One of the TCP flags. SYN stands for synchronize. This flag is used when first establishing a TCP connection and make sure the receiving end knows to examine the sequence number field

SYN_RECEIVED: A TCP socket state that means that a socket previously in a listener state, has received a synchronization request and sent a SYN_ACK back

SYN_SENT: A TCP socket state that means that a synchronization request has been sent, but the connection hasn't been established yet

TCP checksum: A mechanism that makes sure that no data is lost or corrupted during a transfer

TCP segment: A payload section of an IP datagram made up of a TCP header and a data section

TCP window: The range of sequence numbers that might be sent before an acknowledgement is required

URG flag: One of the TCP control flags. URG is short for urgent. A value of one here indicates that the segment is considered urgent and that the urgent pointer field has more data about this

Urgent pointer field: A field used in conjunction with one of the TCP control flags to point out particular segments that might be more important than others

Terms and their definitions from previous modules

A

Address class system: A system which defines how the global IP address space is split up

Address Resolution Protocol (ARP): A protocol used to discover the hardware address of a node with a certain IP address

ARP table: A list of IP addresses and the MAC addresses associated with them

ASN: Autonomous System Number is a number assigned to an individual autonomous system

B

Bit: The smallest representation of data that a computer can understand

Border Gateway Protocol (BGP): A protocol by which routers share data with each other

Broadcast address: A special destination used by an Ethernet broadcast composed by all Fs

Broadcast: A type of Ethernet transmission, sent to every single device on a LAN

C

Cable categories: Groups of cables that are made with the same material. Most network cables used today can be split into two categories, copper and fiber

Cables: Insulated wires that connect different devices to each other allowing data to be transmitted over them

Carrier-Sense Multiple Access with Collision Detection (CSMA/CD): CSMA/CD is used to determine when the communications channels are clear and when the device is free to transmit data

Client: A device that receives data from a server

Collision domain: A network segment where only one device can communicate at a time

Computer networking: The full scope of how computers communicate with each other

Copper cable categories: These categories have different physical characteristics like the number of twists in the pair of copper wires. These are defined as names like category (or cat) 5, 5e, or 6, and how quickly data can be sent across them and how resistant they are to outside interference are all related to the way the twisted pairs inside are arranged

Crosstalk: Crosstalk is when an electrical pulse on one wire is accidentally detected on another wire

Cyclical Redundancy Check (CRC): A mathematical transformation that uses polynomial division to create a number that represents a larger set of data. It is an important concept for data integrity and is used all over computing, not just network transmissions

D

Data packet: An all-encompassing term that represents any single set of binary data being sent across a network link

Datalink layer: The layer in which the first protocols are introduced. This layer is responsible for defining a common way of interpreting signals, so network devices can communicate

Demarcate: To set the boundaries of something

Demarcation point: Where one network or system ends and another one begins

Destination MAC address: The hardware address of the intended recipient that immediately follows the start frame delimiter

Destination network: The column in a routing table that contains a row for each network that the router knows about

DHCP: A technology that assigns an IP address automatically to a new device. It is an application layer protocol that automates the configuration process of hosts on a network

Dotted decimal notation: A format of using dots to separate numbers in a string, such as in an IP address

Duplex communication: A form of communication where information can flow in both directions across a cable

Dynamic IP address: An IP address assigned automatically to a new device through a technology known as Dynamic Host Configuration Protocol

E

Ethernet frame: A highly structured collection of information presented in a specific order

Ethernet: The protocol most widely used to send data across individual links

EtherType field: It follows the Source MAC Address in a dataframe. It's 16 bits long and used to describe the protocol of the contents of the frame

Exterior gateway: Protocols that are used for the exchange of information between independent autonomous systems

F

Fiber cable: Fiber optic cables contain individual optical fibers which are tiny tubes made of glass about the width of a human hair. Unlike copper, which uses electrical voltages, fiber cables use pulses of light to represent the ones and zeros of the underlying data

Five layer model: A model used to explain how network devices communicate. This model has five layers that stack on top of each other: Physical, Data Link, Network, Transport, and Application

Flag field: It is used to indicate if a datagram is allowed to be fragmented, or to indicate that the datagram has already been fragmented

Fragmentation offset field: It contains values used by the receiving end to take all the parts of a fragmented packet and put them back together in the correct order

Fragmentation: The process of taking a single IP datagram and splitting it up into several smaller datagrams

Frame check sequence: It is a 4-byte or 32-bit number that represents a checksum value for the entire frame

Full duplex: The capacity of devices on either side of a networking link to communicate with each other at the exact same time

H

Half-duplex: It means that, while communication is possible in each direction, only one device can be communicating at a time

Header checksum field: A checksum of the contents of the entire IP datagram header

Header length field: A four bit field that declares how long the entire header is. It is almost always 20 bytes in length when dealing with IPv4

Hexadecimal: A way to represent numbers using a numerical base of 16

Hub: It is a physical layer device that broadcasts data to everything computer connected to it

I

IANA: The Internet Assigned Numbers Authority, is a non-profit organization that helps manage things like IP address allocation

Identification field: It is a 16-bit number that's used to group messages together

Interface: For a router, the port where a router connects to a network. A router gives and receives data through its interfaces. These are also used as part of the routing table

Interior gateway: Interior gateway protocols are used by routers to share information within a single autonomous system

Internet Protocol (IP): The most common protocol used in the network layer

Internet Service Provider (ISP): A company that provides a consumer an internet connection

Internetwork: A collection of networks connected together through routers - the most famous of these being the Internet

IP datagram: A highly structured series of fields that are strictly defined

IP options field: An optional field and is used to set special characteristics for datagrams primarily used for testing purposes

L

Line coding: Modulation used for computer networks

Local Area Network (LAN): A single network in which multiple devices are connected

M

MAC(Media Access Control) address: A globally unique identifier attached to an individual network interface. It's a 48-bit number normally represented by six groupings of two hexadecimal numbers

Modulation: A way of varying the voltage of a constant electrical charge moving across a standard copper network cable

Multicast frame: If the least significant bit in the first octet of a destination address is set to one, it means you're dealing with a multicast frame. A multicast frame is similarly set to all devices on the local network signal, and it will be accepted or discarded by each device depending on criteria aside from their own hardware MAC address

N

Network Address Translation (NAT): A mitigation tool that lets organizations use one public IP address and many private IP addresses within the network

Network layer: It's the layer that allows different networks to communicate with each other through devices known as routers. It is responsible for getting data delivered across a collection of networks

Network port: The physical connector to be able to connect a device to the network. This may be attached directly to a device on a computer network, or could also be located on a wall or on a patch panel

Network switch: It is a level 2 or data link device that can connect to many devices so they can communicate. It can inspect the contents of the Ethernet protocol data being sent around the network, determine which system the data is intended for and then only send that data to that one system

Next hop: The IP address of the next router that should receive data intended for the destination networking question or this could just state the network is directly connected and that there aren't any additional hops needed. Defined as part of the routing table

Node: Any device connected to a network. On most networks, each node will typically act as a server or a client

Non-routable address space: They are ranges of IPs set aside for use by anyone that cannot be routed to

O

Octet: Any number that can be represented by 8 bits

Organizationally Unique Identifier (OUI): The first three octets of a MAC address

OSI model: A model used to define how network devices communicate. This model has seven layers that stack on top of each other: Physical, Data Link, Network, Transport, Session, Presentation, and Application

P

Padding field: A series of zeros used to ensure the header is the correct total size

Patch panel: A device containing many physical network ports

Payload: The actual data being transported, which is everything that isn't a header

Physical layer: It represents the physical devices that interconnect computers

Preamble: The first part of an Ethernet frame, it is 8 bytes or 64 bits long and can itself be split into two sections

Protocol field: A protocol field is an 8-bit field that contains data about what transport layer protocol is being used

Protocol: A defined set of standards that computers must follow in order to communicate properly is called a protocol

R

Router: A device that knows how to forward data between independent networks

Routing protocols: Special protocols the routers use to speak to each other in order to share what information they might have

S

Server: A device that provides data to another device that is requesting that data, also known as a client

Service type field: An eight bit field that can be used to specify details about quality of service or QoS technologies

Simplex communication: A form of data communication that only goes in one direction across a cable

Source MAC address: The hardware address of the device that sent the ethernet frame or data packet. In the data packet it follows the destination MAC address

Start Frame Delimiter (SFD): The last byte in the preamble, that signals to a receiving device that the preamble is over and that the actual frame contents will now follow

Static IP address: An IP address that must be manually configured on a node

Subnet mask: 32-bit numbers that are normally written as four octets of decimal numbers

Subnetting: The process of taking a large network and splitting it up into many individual smaller sub networks or subnets

T

Time-To-Live field (TTL): An 8-bit field that indicates how many router hops a datagram can traverse before it's thrown away

Total hops: The total number of devices data passes through to get from its source to its destination. Routers try to choose the shortest path, so fewest hops possible. The routing table is used to keep track of this

Total length field: A 16-bit field that indicates the total length of the IP datagram it's attached to

Transmission Control Protocol (TCP): The data transfer protocol most commonly used in the fourth layer. This protocol requires an established connection between the client and server

Transport layer: The network layer that sorts out which client and server programs are supposed to get the data

Twisted pair cable: The most common type of cabling used for connecting computing devices. It features pairs of copper wires that are twisted together

U

Unicast transmission: A unicast transmission is always meant for just one receiving address

User Datagram Protocol (UDP): A transfer protocol that does not rely on connections. This protocol does not support the concept of an acknowledgement. With UDP, you just set a destination port and send the data packet

V

Virtual LAN (VLAN): It is a technique that lets you have multiple logical LANs operating on the same physical equipment

VLAN header: A piece of data that indicates what the frame itself is. In a data packet it is followed by the EtherType
